

Shahriar Sakib

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PROFILE

Final-year CSE student at BRAC University with hands-on experience in machine learning, full-stack development, and IoT systems. Currently conducting thesis research on bio-inspired swarm routing algorithms for EV emergency coordination in urban networks. Seeking a software engineering or ML engineering role where applied research and production systems meet.

EDUCATION

B.Sc. in Computer Science & Engineering	2022 – 2026
BRAC University, Dhaka · CGPA: 3.45 / 4.00	
Higher Secondary Certificate, Science	2020
Dhaka City College, Dhaka · GPA: 5.00 / 5.00	
Secondary School Certificate, Science	2018
Govt. Laboratory High School, Dhaka · GPA: 5.00 / 5.00	

TECHNICAL SKILLS

Languages	Python, SQL
ML / Data	NumPy, Pandas, Matplotlib, Scikit-learn
Deep Learning	TensorFlow / Keras · CNN, RNN, LSTM, GRU, Transformer
Web	React, Node.js, Express, Socket.IO, Laravel, PHP
Databases	MySQL, MongoDB Atlas, Redis (Upstash)
Tools	Git, Jupyter Notebook, VS Code, Overleaf, SUMO, Vercel
Interests	Machine Learning, NLP, Computer Vision, IoT, Swarm AI

UNDERGRADUATE THESIS

E³-Hybrid Swarm Routing for EV Emergency Coordination	2025 – 2026
Supervisor: Dr. Md. Khalilur Rahman, BRAC University · Status: In Progress	
- Proposes E³-Hybrid, a novel algorithm combining ACO, PSO, and BCO for decentralised multi-agent EV routing under emergency conditions in urban environments. - Runs 1,050 simulation batches (7 algorithms × 6 stress scenarios × 25 seeds) on a SUMO-based Midtown Manhattan network with full statistical analysis and IEEE-format visualisations.	

PROJECTS

ConnectHub: Real-Time Messaging Application  Live Demo	2025
MongoDB Atlas, Express.js, React.js, Node.js · Socket.IO · Redis (Upstash)	

- Built a production-grade real-time chat application with WebSocket-based messaging via Socket.IO, deployed live on Vercel.
- Integrated Upstash Redis for in-memory caching to reduce database load and improve message delivery latency.
- Implemented user authentication, persistent message history with MongoDB Atlas, and a responsive UI.

Doctor Handwriting Recognition

2026

Python, TensorFlow/Keras · DenseNet121, MobileNetV3, ResNet50, ViT-B16, CNN-Transformer

- Benchmarked five deep learning architectures for physician prescription handwriting classification.
- Designed the full pipeline: image preprocessing, training, evaluation, and cross-model comparison.

News Headline Text Classification

2025

Python, Scikit-learn, TensorFlow · SimpleRNN, LSTM, GRU, Logistic Regression, DNN

- Evaluated 24 configurations across preprocessing strategies, text representations, and model types.
- Achieved best F1 of 0.9276 with Bidirectional GRU; documented findings in an IEEE-format report.

E-Commerce Platform

2025

MongoDB, Express, React, Node.js (MERN)

- Built a full-stack e-commerce application with product listings, JWT authentication, and cart functionality.

A.Q.U.A.: Aquarium Automation System

2025

Arduino Uno R3, ESP8266, IoT Sensors, Embedded C

- Developed an IoT-enabled automation system with scheduled feeding, water management, and WiFi control.

Partner Matching Web Platform

2024

Laravel, PHP, MySQL

- Built a compatibility-based matching application with user profiling and a relational database backend.

LANGUAGES

Bangla	Native
English	Proficient (Academic & Professional)